

Thesis Proposal – Reinforcement Learning Environment for Collectible Card Game Battling



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1 Introduction

Games have long served AI research as a treasure trove of simulations providing varied challenges with relatively well-established rules, tractable scopes of required skills and abundance of human expertise. After deep-learning Monte-Carlo-Tree-Search approaches achieved unambiguous super-human play in go (and similar symmetric perfect information games; shogi and chess), research into structurally complex games has intensified.[1] Over the last few years collectible card games (CCG) have been the focus of various works due to several such structural complexities:

1. enormous state space [2]
2. complex rules defined in natural language [3]
3. non trivial state transitions [4]
4. structurally different but connected game stages [5]
 - a) Deckbuilding
 - b) Battling

While several teams have shown that, in simple CCGs, self learned agents can outperform expert-crafted classical approaches during Drafting, few works exist on deep-learning for Battling. From personal experience in several CCGs we believe that, in any sufficiently complex CCG, Battling, Drafting and Deck-Building are linked strongly enough that comprehensive research on either has to ultimately involve the other. Research on Battling thus not only provides a new set of challenges but also enables more informed analysis of previous work on Drafting and consequently is essential for the development of a universal CCG-playing agent.

Current implementations of CCG engines are usually high-fidelity, resource intense implementations intended for human use (or sometimes classical AI) without a proper API for AI training. A notable exception exists in LOCM [6] for which an openai gym API, capable of running multiple games a second, is implemented. While fundamental to research on CCG AI in the last decade LOCM eschews multiple sources of structural complexity to improve accessibility for various self learning approaches and execution speed. It consequently also deviates from environments where human expertise is most abundant, namely preexisting CCGs.

We believe that structural complexity and human expertise are essential to improve CCG agents in the long term, thus we propose to implement an CCG environment which offers an efficiently computable, simplified (but congruent) version of a preexisting CCG. The focus of the proposed implementation shall be on computationally inexpensive sources of structural complexity as well as expandability to enable a gradual approach to the reference CCG. Due to its high-complexity even among CCGs[7], its wide spread among players and its lacking representation in Reinforcement learning, we choose Magic: The Gathering as our reference CCG.

2 State of Research

CCG research encompasses static analysis (game complexity or card balance) as well as agents for playing. Deep learning approaches of the last few years have primarily focused on Deckbuilding, relying on relatively simple baselines for Battling. In general such agents are trained for LOCM and do not compete with human but rather with each other. A notable exception is the work of Świechowski et al.[8] who report human level battling for Hearthstone. Wei Xi et al. also stand out as they consider the joint problem of Deckbuilding

and Battling, which is unusual at the time of writing. Overall we identify the following currently unaddressed problems in CCG agent research:

1. Low Interpretability
 - a) Most teams track only winrates
 - b) Low/No human play makes it hard to determine if agents are learning degenerate/exploitable play patterns[9]
 - c) No focus on extracting (human-readable) knowledge from trained agents
2. LoCM centricity
 - a) LoCM falls on the low end of CCG complexity
 - b) Engines for more complex CCGs are computationally expensive and do not provide standard APIs

3 Proposed Method

To adress the aforementioned shortcoming we propose to implement a M:TG-inspired CCG environment with the following properties:

1. Provides new and complex challenges compared to the LoCM [7]
 - a) Complex Combat: Declared Combat
 - b) Complex Resource System: Resource Cards
 - c) Redraw: M:TG-style Mulligan
2. Is of similar computational complexity as LoCM: tens of games per second on a personal pc
3. Provides a standard RL API: Implements the OpenAiGym API
4. Enables Human-Agent "Crossplay"
5. Enables evaluation of expert-defined metrics beyond winrate

References

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